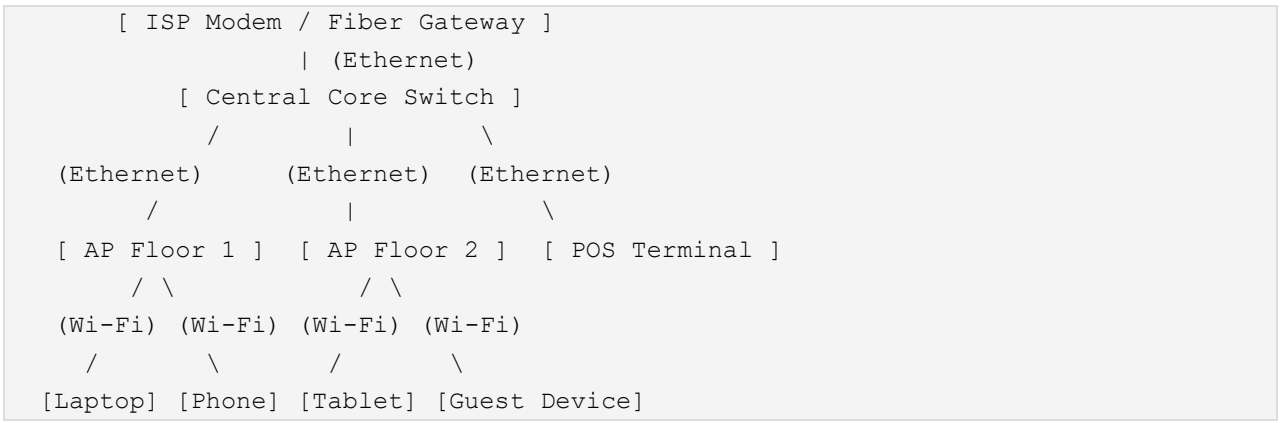
# NETWORK TOPOLOGIES & DESIGN

**1. Draw a network topology diagram for a small café Wi-Fi setup using the star topology, labeling all devices (router, access points, user devices) and connections clearly.**

**Answer & Topological Description:**

In a star topology café setup, all peripheral devices connect directly to a central hub or switch:




**Description:** The ISP modem feeds into a core managed switch. The wireless Access Points (APs) and Point-of-Sale (POS) terminals connect via wired Ethernet cables radiating outward from the central switch, forming a classic Star Topology.

**2. Design a network for a two-floor co-working space (like a mini WeWork) using a ringtopology for each floor and explain how devices are connected to maintain communication between floors.**

**Answer & Design Explanation:**

**Floor-wise Ring Topology:** Each floor operates a dual-fiber or token ring network where every workstation connects to its neighbor in a closed loop, ensuring data circulates around the ring.

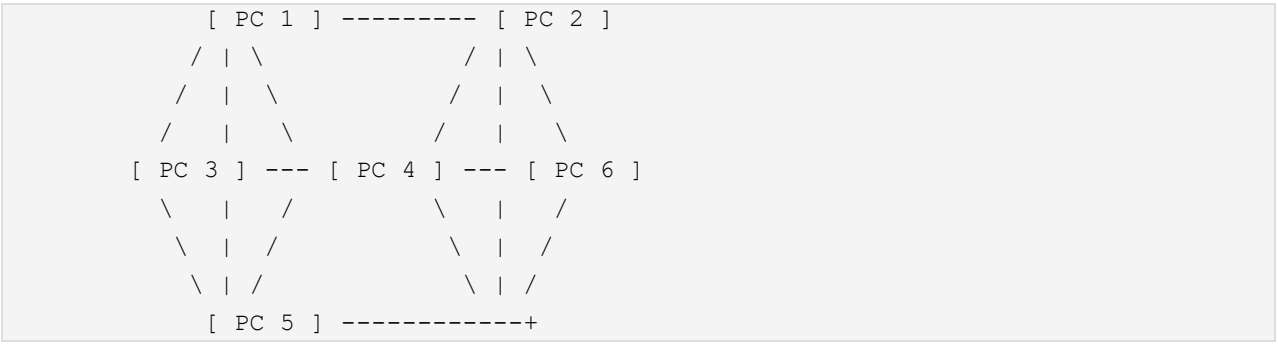
**Inter-Floor Communication:** Each floor's ring is anchored by a floor gateway router or multi-station access unit (MAU). The Floor 1 ring switch connects to the Floor 2 ring switch via a dedicated highspeed Gigabit uplink backbone connection.

**Fault Tolerance:** If a cable or device fails on Floor 1, dual-ring counter-rotation automatically loops the data back, isolating the fault without disrupting Floor 2 operations or inter-floor communication.

**3. Create a mesh topology diagram for a gaming café with 6 PCs, showing how redundancyis achieved if any one PC or cable fails.**

**Answer & Topological Layout:**

In a full mesh topology among 6 gaming PCs, every PC has a direct dedicated link to every other PC (5 connections per PC, totaling 15 unique interconnects):




**Redundancy Explanation:** If the direct cable between PC 1 and PC 2 fails, traffic is instantly re-routed via alternative paths (e.g., PC 1 -> PC 3 -> PC 2), ensuring zero downtime during competitive multiplayer gaming sessions.

**4. Plan a multi-site network for a food delivery company with 3 city offices (Ahmedabad,Surat, Vadodara) — show how you would connect the sites, what topology you would use between them, and briefly explain your choice.**

**Answer & Multi-Site Plan:**

**Topology Choice:** A **Partial Mesh (Hub-and-Spoke with redundancy)** WAN topology connecting Ahmedabad (Headquarters), Surat, and Vadodara.

**Inter-Site Connections:**

- Ahmedabad (HQ) acts as the central hub connecting to both Surat and Vadodara via high-speed MPLS / IPsec VPN tunnels over fiber broadband.
- A direct redundant backup link connects Surat and Vadodara directly.

**Justification:** This topology optimizes cost while ensuring high availability. If the primary Ahmedabad hub experiences an outage, Surat and Vadodara can communicate through the backup cross-link, maintaining regional dispatch operations for the delivery network.

**5. Imagine your network needs to double in size next year (e.g., from 10 to 20 devices persite). Write a short expansion plan describing what changes you would make to your original topology to ensure scalability and minimal downtime.**

**Answer & Scalability Expansion Plan:**

1. **Architectural Transition:** Upgrade the single-switch star topology to a **Hierarchical Star Topology** (Access layer switches cascading into a Core distribution switch).
2. **Hardware Upgrades:** Replace 10-port switches with modular 24/48-port Gigabit managed switches and deploy high-capacity enterprise routers at branch sites.
3. **Network Segmentation (VLANs):** Implement VLANs to logically segregate new devices (e.g., IoT, guest Wi-Fi, staff terminals), reducing broadcast domain congestion.
4. **Minimal Downtime Strategy:** Perform core infrastructure upgrades during scheduled maintenance windows (late night), pre-configure new switches, and use link aggregation (EtherChannel) to expand bandwidth seamlessly without interrupting active traffic.